

# CAD RISK PREDICTION SYSTEM

with India-Specific Genetic Risk & Digital Twin Component

## FINAL PIPELINE BLUEPRINT & METHODOLOGY DOCUMENT

April 2026 | Capstone Project

| PROJECT SNAPSHOT |                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Datasets         | 70k Cardio (lifestyle), 1,190 Clinical (Cleveland+Hungary+Statlog), PGS2809 weights, GenomeIndia freqs                                                                          |
| PRS Approach     | Population-level PRS (Path A) — Indian-ancestry expected PRS using PGS2809 weights + GenomeIndia allele frequencies. Monte Carlo (10k iterations) gives 95% CI on the estimate. |
| Digital Twin     | ML-based counterfactual engine (PRIMARY) + PulsePhysio integration (OPTIONAL/conditional)                                                                                       |
| Fusion Strategy  | Stacked meta-learner (logistic regression on out-of-fold predictions)                                                                                                           |
| Explainability   | SHAP TreeExplainer with domain grouping (Genetic   Lifestyle   Clinical)                                                                                                        |
| Target Venue     | Peer-reviewed journal — requires AUC, NRI, IDI, calibration, ablation, and DT validation                                                                                        |

# 0. Scope Definition — What We Are and Are Not Building

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## 0.1 IN SCOPE (what we are doing)

### Core ML Pipeline

- Preprocessing and cleaning of all three data sources independently
- Training separate ML models on lifestyle (70k) and clinical (1,190) datasets
- Population-level PRS Score computation using PGS2809 weights and GenomeIndia Indian-ancestry allele frequencies
- Stacked fusion meta-learner combining genetic, lifestyle, and clinical risk
- SHAP-based explainability with domain grouping (Genetic / Lifestyle / Clinical breakdown)
- Counterfactual intervention engine — the primary digital twin mechanism
- Risk stratification bands (Low / Moderate / High / Very High)
- Ablation study validating fusion over individual components
- Full validation suite: AUC-ROC, Brier score, NRI, IDI vs Framingham
- Clinical sanity check battery and literature benchmarking for the DT
- Paper-quality figures for publication submission

### PulsePhysio Integration (OPTIONAL — conditional on scope approval)

- Initialization of Pulse patient models from Cleveland/Statlog clinical features
- Exercise scenario simulation (ExerciseAction) to extract haemodynamic deltas
- Smoking cessation and weight loss haemodynamic scenarios
- Mapping Pulse outputs (new BP, HR) back into the ML feature vector for re-scoring
- This is included as a modular add-on that can be enabled or removed without breaking the core pipeline
- If scope cannot be reduced to ML-only DT: use Pulse. If permission granted to skip: use ML DT only.

## 0.2 OUT OF SCOPE (explicitly excluded — do not attempt)

### Excluded from this project

- Individual patient-level genotyping / SNP data — we do NOT have this; do not claim to
- True longitudinal trajectory model — we have cross-sectional data only; trajectory is a SIMULATED PROJECTION, not a trained temporal model
- Mendelian Randomisation or any formal causal inference machinery
- SNP-to-physiology chain modelling (SNP → LDL → atherosclerosis → BP pathway)
- BioGears — wrong tool for this use case (designed for acute pharmacological simulation, not population-level CAD risk)
- Wearable integration — future work only, do not include in core model
- Regulatory submission, EHR integration, or clinical deployment
- Individual-level genotype PRS (Path B) — we are doing Path A: population-level expected PRS only
- Direct-to-doctor product — not our launch market
- Any claim of 'causal' inference — use 'feature attribution' and 'counterfactual intervention analysis' throughout

# 1. Data Inventory — What We Have and What Each Dataset Does

## 1.1 The Three Data Streams

Every pipeline step draws from one or more of these three streams. Understanding their roles, limitations, and how they connect is the first thing to get right.

| Dataset                                           | Size                        | Schema                                                                                                | Role in Pipeline                                                                                                   | Key Limitation                                                                                                                                                                        |
|---------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cardio_Data.csv<br>(lifestyle/70k)                | 70,000 rows                 | age, gender, height, weight, ap_hi, ap_lo, cholesterol (cat), gluc (cat), smoke, alco, active, target | PRIMARY training data for lifestyle+clinical ML model (Model B)                                                    | Lifestyle features are binary flags only (smoke, alco, active) — not continuous. Cholesterol and glucose are ordinal categories (1/2/3), not mg/dL values. Age is in days, not years. |
| Cleveland + Hungary + Statlog combined<br>(1,190) | 1,190 rows<br>(303+294+593) | age, sex, cp, trestbps, chol (mg/dL), fbs, restecg, thalach, exang, oldpeak, slope, ca, thal, target  | VALIDATION dataset. Also used to initialize PulsePhysio (continuous clinical values required). Clinical benchmark. | Small N. Some missing values in ca and thal. Not an Indian cohort — used for methodological validation, not Indian-population claims.                                                 |
| PGS2809<br>(pgs_catalog_2809.tsv)                 | 182 SNPs                    | rsID, chr_name, chr_position, effect_allele, effect_weight (beta)                                     | PRS weight file. Defines which SNPs contribute to CAD risk and by how much.                                        | Weights alone are meaningless without population allele frequencies. Must be combined with GenomeIndia.                                                                               |
| GenomeIndia (9768 GI Summary Stats, chr1-22)      | 129.9M variants             | Chromosome, Position, ID, Ref allele, Alt allele, Alt allele frequency                                | Indian-ancestry allele frequencies. Used to compute expected dosage per SNP for population-level PRS.              | Population-level only. We do not have individual patient genotypes.                                                                                                                   |

## 1.2 The PRS Terminology Clarification

### IMPORTANT — PRS vs PGS, Path A vs Path B

PGS2809 is the name of the polygenic score in the PGS Catalog (a database of validated genetic risk scores).

PRS (Polygenic Risk Score) is the computed value for an individual or population using PGS weights. We are computing a POPULATION-LEVEL PRS (Path A): using GenomeIndia Indian-ancestry allele frequencies + PGS2809 beta weights to compute the EXPECTED PRS for an Indian individual under Hardy-Weinberg equilibrium.

We are NOT computing individual patient PRS because we do not have individual patient genotype data (SNP measurements per patient).

The population-level PRS scalar represents the Indian-ancestry genetic baseline risk. It is used as a FIXED OFFSET that shifts baseline risk upward or downward for the Indian population relative to European-calibrated risk calculators.

In the paper: frame this as 'India-specific polygenic risk score calibration using GenomeIndia population allele frequencies.'



## 2. Pipeline Architecture — End-to-End Overview

The pipeline has eight stages that flow sequentially. Each stage produces outputs that feed the next. The diagram below shows the logical flow; each stage is described in detail in Section 3.

| Stage                               | Notebook(s)                        | What Happens                                                                              | Input From                      | Output To                                           |
|-------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------|
| Stage 1: Preprocessing              | NB1 (70k), NB2 (1190)              | Clean, encode, split both datasets independently                                          | Raw CSVs                        | Stage 2, 3, 4                                       |
| Stage 2: Genome Preprocessing       | NB3                                | Extract GenomeIndia frequencies for PGS2809 SNPs                                          | GenomeIndia TSVs + PGS2809 TSV  | Stage 3                                             |
| Stage 3: PRS Computation            | NB4                                | Compute population-level PRS scalar + Monte Carlo 95% CI + per-SNP contribution breakdown | Harmonised genetic map from NB3 | Stage 5 (genetic feature — PRS mean used as offset) |
| Stage 4: Model Training — Lifestyle | NB5                                | Train 6 ML models on 70k dataset, select best                                             | NB1 train/test splits           | Stage 6                                             |
| Stage 5: Model Training — Clinical  | NB6                                | Train 6 ML models on 1190 dataset, select best                                            | NB2 train/test splits           | Stage 6                                             |
| Stage 6: Genetic Risk Integration   | NB7                                | Integrate PRS into both models, risk stratification                                       | NB4 PRS scalar + NB5/NB6 models | Stage 7                                             |
| Stage 7: Fusion + Explainability    | NB8                                | Stacked meta-learner fusion, SHAP attribution                                             | NB7 integrated scores           | Stage 8                                             |
| Stage 8: Digital Twin + Validation  | NB9 (ML DT), NB10 optional (Pulse) | Counterfactual engine, trajectory viz, full validation suite, paper figures               | NB8 fused model                 | FINAL OUTPUTS                                       |

### Architecture in one sentence

Raw data (70k lifestyle + 1190 clinical + PGS2809 weights + GenomeIndia frequencies) flows through independent preprocessing → a population-level PRS scalar is computed and integrated into both trained models as a fixed genetic offset → a stacked meta-learner fuses the model outputs → SHAP attribution breaks down risk by domain → a counterfactual engine simulates lifestyle interventions → a simulated trajectory visualization and full validation suite complete the system.

### 3. Notebook-by-Notebook Specification

There are 10 notebooks total. NB1 and NB2 can be run in parallel. NB5 and NB6 can be run in parallel after their prerequisites. NB10 (PulsePhysio) is conditional.

| NB1: preprocessing_70k.ipynb |                                                                                                                                 |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| PURPOSE                      | Clean, transform, and split the 70,000-row cardiovascular lifestyle dataset into train/test sets ready for model training.      |
| INPUTS                       | Cardio_Data.csv (raw, 70,000 rows, 12 columns)                                                                                  |
| OUTPUTS                      | outputs/lifestyle/df_lifestyle_train.csv   outputs/lifestyle/df_lifestyle_test.csv   outputs/lifestyle/preprocessing_report.txt |

#### NB1 — What to do inside, step by step

- Section 1 — Load and Inspect: Load Cardio\_Data.csv. Print shape, dtypes, .describe(), missing value counts, and class balance of 'target' column.
- Section 2 — Age conversion: 'age' is stored in days. Convert to years:  $\text{age\_years} = \text{age} / 365.25$ . Drop the original 'age' column. This is a known quirk of this specific dataset.
- Section 3 — Gender encoding: 'gender' is a string ('male'/'female'). Encode as binary: male=1, female=0. Rename to 'sex' for consistency with the clinical dataset.
- Section 4 — Blood pressure sanity: ap\_hi (systolic) and ap\_lo (diastolic) can contain physiologically impossible values (e.g.,  $\text{ap\_hi} < 50$  or  $> 250$ ,  $\text{ap\_lo} < 30$  or  $> 200$ , or  $\text{ap\_hi} \leq \text{ap\_lo}$ ). Remove or clip these outliers. Document the cutoffs used.
- Section 5 — Cholesterol and glucose: These are ordinal categories (1=normal, 2=above normal, 3=well above normal) in this dataset. Do NOT treat as continuous. Keep as ordinal-encoded integers. Note explicitly in the paper that these are not mg/dL values — this is a data limitation.
- Section 6 — BMI derivation: Compute  $\text{BMI} = \text{weight} / (\text{height}/100)^2$ . Add as a new feature. Check for implausible BMI values ( $< 10$  or  $> 70$ ).
- Section 7 — Class imbalance check: Compute class balance ratio. If ratio exceeds 1.5:1, apply SMOTE on the training set ONLY after splitting. Never SMOTE the test set.
- Section 8 — Train/test split: 70% train, 15% val, 15% test using stratified split (preserves class ratio). Fix random\_state=42 throughout.
- Section 9 — Standardisation: Fit StandardScaler on TRAINING set only. Save the scaler object. Apply to val and test sets using training parameters.
- Section 10 — Export: Save train and test CSVs. Save the scaler as a .pkl file. Write a preprocessing report documenting all transformations and dropped rows.

#### NB1 Data Limitation — Must Acknowledge in Paper

The 70k dataset's lifestyle features (smoke, alco, active) are binary presence/absence flags, not continuous measures like pack-years, units/week, or MET-min/week. This limits the precision of counterfactual lifestyle simulation. The paper should state: 'Due to data availability constraints, lifestyle factors in the training dataset are represented as binary indicators rather than continuous measures; this limits the granularity of intervention dose-response estimation.'

#### NB2: preprocessing\_1190\_clinical.ipynb

|                |                                                                                                                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PURPOSE</b> | Clean and merge the three clinical datasets. This dataset has richer continuous clinical features (actual mg/dL cholesterol, mmHg BP, thalach, etc.) and will be used for clinical model training and PulsePhysio initialization. |
| <b>INPUTS</b>  | Cleveland_Data.csv (303 rows) + Hungarian_Data.csv (294 rows) + heart_statlog_cleveland_hungary_final.csv (1,190 rows)                                                                                                            |
| <b>OUTPUTS</b> | outputs/clinical/df_clinical_train.csv   outputs/clinical/df_clinical_test.csv   outputs/clinical/preprocessing_report.txt                                                                                                        |

## NB2 — What to do inside

- Section 1 — Load all three: Inspect each individually. Note schema differences. The combined heart\_statlog file (1,190) is the merged version; prefer it as the primary source to avoid double-counting.
- Section 2 — Schema harmonisation: Ensure column names are consistent across sources. The combined file already does this but verify. Rename columns to a standard schema: age, sex, chest\_pain\_type, resting\_bp, cholesterol, fasting\_bs, resting\_ecg, max\_hr, exercise\_angina, oldpeak, st\_slope, target.
- Section 3 — Missing values: 'ca' (number of major vessels) and 'thal' have missing values in the Cleveland subset. Impute with median (preferred over mean for skewed clinical variables) or drop rows where both are missing. Document your choice.
- Section 4 — Outlier check: Resting BP < 80 or > 220, cholesterol = 0 (known artifact in this dataset — some Cleveland records have chol=0, which is physiologically impossible; drop these rows), max\_hr < 60 or > 220.
- Section 5 — Class balance and SMOTE: Same logic as NB1. Apply if needed.
- Section 6 — Train/test split: 70/15/15 stratified. Same random\_state=42.
- Section 7 — Standardise, save scaler, export CSVs.
- Section 8 — Preserve a 'raw clinical' version of the test set (unscaled, with original mmHg and mg/dL values) specifically for PulsePhysio initialization. Save as outputs/clinical/df\_clinical\_test\_raw.csv.

| NB3: genome_data_preprocessing.ipynb |                                                                                                                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PURPOSE</b>                       | Extract Indian-ancestry allele frequencies from GenomeIndia for each SNP in PGS2809. This is the critical linking step that makes this system India-specific.                                                |
| <b>INPUTS</b>                        | pgs_catalog_2809.tsv (182 SNPs with rsID, chr, position, effect_allele, effect_weight) + GenomeIndia TSV files GI_9768_CBR-NIBMG_JointCall_AF_chr1.tsv through chr22.tsv (allele frequencies per chromosome) |
| <b>OUTPUTS</b>                       | outputs/genetics/harmonized_genetic_map.xlsx — one row per SNP with columns: rsID, chr, position, effect_allele, effect_weight, indian_ancestry_risk_allele_freq, snp_status                                 |

## NB3 — What to do inside

- Section 1 — Load PGS2809: Load pgs\_catalog\_2809.tsv. Print all 182 SNPs. Confirm columns: rsID, chr\_name, chr\_position, effect\_allele, effect\_weight.
- Section 2 — Load GenomeIndia by chromosome: For each unique chromosome in PGS2809 (not all 22 are represented), load the corresponding GenomeIndia TSV. These files are large — load only the chromosome files you need. Match by genomic position (chr + position).

- Section 3 — Position matching: Match each PGS2809 SNP to its GenomeIndia record by chromosome number and base pair position. The GenomeIndia files include both the Reference (REF) and Alternate (ALT) allele plus ALT allele frequency.
- Section 4 — Allele alignment: Check whether the effect\_allele from PGS2809 matches the REF or ALT allele in GenomeIndia. If effect\_allele matches ALT: indian\_freq = ALT\_freq. If effect\_allele matches REF: indian\_freq = 1 - ALT\_freq (complement). If no match: mark SNP as 'ambiguous' or 'missing'.
- Section 5 — Filtering and logging: Identify SNPs that did not match (position not found in GenomeIndia). These are likely rare variants absent from the Indian cohort. Log them. The blueprint previously noted 2 SNPs were excluded — document your exact exclusion count and reasons.
- Section 6 — Harmonised map: Create final DataFrame with columns: rsID, chr\_name, chr\_position, effect\_allele, effect\_weight (beta), indian\_ancestry\_risk\_allele\_freq, snp\_status (matched/complemented/excluded). Save as Excel for human readability.

#### NB4: prs\_score\_computation.ipynb

|         |                                                                                                                                                                                                                                                                                                                                                     |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PURPOSE | Compute the population-level PRS for Indian ancestry using PGS2809 weights and GenomeIndia allele frequencies. Includes a Monte Carlo uncertainty quantification step that converts the single PRS scalar into a distribution with a 95% confidence interval — this is the methodologically rigorous way to report a population-level PRS estimate. |
| INPUTS  | outputs/genetics/harmonized_genetic_map.xlsx (from NB3)                                                                                                                                                                                                                                                                                             |
| OUTPUTS | outputs/genetics/prs_population_score.csv (scalar + 95% CI from Monte Carlo)  <br>outputs/genetics/per_snp_contribution.csv  <br>outputs/genetics/prs_monte_carlo_distribution.csv  <br>outputs/genetics/prs_feature_vector.pkl  <br>outputs/figures/prs_monte_carlo_distribution.png  <br>outputs/figures/snp_contribution_bar.png                 |

#### NB4 — Naming clarification: PGS vs PRS

##### PGS2809 vs PRS — what these terms mean

PGS2809 is the NAME of the polygenic score entry in the PGS Catalog (an international database). The file pgs\_catalog\_2809.tsv is its actual filename — keep this filename as-is.

PRS (Polygenic Risk Score) is the VALUE we COMPUTE using PGS2809's weights applied to our population's allele frequencies.

Correct usage: 'We computed a population-level PRS for Indian ancestry using SNP weights from PGS2809 and allele frequencies from the GenomeIndia dataset.'

In this notebook and all downstream notebooks: the computed value is always called 'PRS'. The input file reference is 'PGS2809 weights'. Never call the computed output a 'PGS score'.

#### NB4 — Core PRS computation

Under Hardy-Weinberg equilibrium, the expected allelic dosage for a diploid individual at a given SNP is  $2 \times \text{allele\_frequency}$ . The expected contribution of each SNP to the total PRS is therefore:

$\text{per\_snp\_contribution} = 2 \times \text{indian\_ancestry\_risk\_allele\_freq} \times \text{effect\_weight\_beta}$

$\text{population\_PRS} = \text{sum}(\text{per\_snp\_contribution})$  across all retained SNPs

- Section 1 — Load harmonised map, check for nulls, print retained SNP count.



- Section 2 — Compute per-SNP contributions using the formula above. Sum to get the point-estimate population PRS scalar.
- Section 3 — Top SNP analysis: Plot bar chart of per-SNP contributions sorted by absolute magnitude. Identify top 10 contributors by rsID. Annotate with chromosomal region. This becomes a paper figure.

## NB4 — Monte Carlo Uncertainty Quantification (Section 4) — WHY and HOW

The simple sensitivity analysis (vary  $\pm 10\%$ ) gives a rough bound but is not statistically principled. The problem is that GenomeIndia allele frequencies are themselves ESTIMATES from 9,768 samples — they have sampling uncertainty. A single allele frequency of 0.35 estimated from 9,768 individuals has a standard error of approximately  $\sqrt{0.35 \times 0.65 / 9768} \approx 0.0048$ . Monte Carlo propagates this uncertainty through all 182 SNPs simultaneously and gives you a proper confidence interval on the PRS.

### Why Monte Carlo here is the right call for publication

Reviewers of genomics and computational medicine papers will ask: 'How sensitive is your PRS to uncertainty in the input allele frequencies?' A  $\pm 10\%$  fixed sensitivity sweep is a crude answer. A Monte Carlo-derived 95% CI is a rigorous answer.

It converts your single PRS scalar into 'Indian-ancestry expected PRS =  $X \pm Y$  (95% CI: [A, B])' — this is how PRS uncertainty is reported in the literature (e.g., Khera et al. 2018 supplements).

It costs almost nothing computationally: 10,000 iterations on 182 SNPs runs in seconds on any laptop.

It strengthens the methods section without adding complexity to the rest of the pipeline — the mean of the Monte Carlo distribution is what gets used downstream as the PRS scalar. The CI is reported in the paper.

## Monte Carlo implementation — step by step

1. For each SNP  $i$  ( $i = 1$  to 182), model the allele frequency as a Beta distribution:  $\text{freq}_i \sim \text{Beta}(\alpha_i, \beta_i)$ , where  $\alpha_i = \text{observed\_freq} \times N_{\text{samples}}$  and  $\beta_i = (1 - \text{observed\_freq}) \times N_{\text{samples}}$ . For GenomeIndia,  $N_{\text{samples}} = 9768$ .
2. Run 10,000 Monte Carlo iterations. In each iteration: sample one allele frequency per SNP from its Beta distribution, compute  $\text{per\_snp\_contribution} = 2 \times \text{sampled\_freq} \times \text{effect\_weight}$ , sum all contributions to get one PRS sample.
3. After 10,000 iterations: compute mean, standard deviation, 2.5th percentile, and 97.5th percentile of the PRS distribution. The mean is your PRS scalar for downstream use. The [2.5th, 97.5th] percentile range is your 95% CI.
4. Plot: histogram of the 10,000 PRS samples with the mean and 95% CI marked. This is a publication-ready figure.
5. Identify which SNPs contribute most to variance in the PRS distribution (variance =  $(2 \times \text{freq} \times (1 - \text{freq}) / N_{\text{samples}}) \times \beta^2$  per SNP). The top variance-contributing SNPs are where GenomeIndia frequency uncertainty matters most — mention this in the paper.

### What Monte Carlo is NOT used for in this project

Monte Carlo for 'virtual genome generation' (generating thousands of synthetic patients for population-level insights) is OUT OF SCOPE. That is a public health analytics application, not a clinical risk prediction paper contribution. Do not include it.

Monte Carlo for simulation of patient trajectories is also out of scope — that would require longitudinal data we do not have.

The ONLY Monte Carlo in this project is: allele frequency uncertainty propagation in NB4, producing the 95% CI on the Indian-ancestry population PRS. That is it.

- Section 5 — Export: prs\_population\_score.csv (with columns: prs\_mean, prs\_std, prs\_ci\_lower, prs\_ci\_upper), per\_snp\_contribution.csv, prs\_monte\_carlo\_distribution.csv (all 10,000 samples), prs\_feature\_vector.pkl (mean value for downstream use).

### NB5: model\_training\_lifestyle.ipynb

|                |                                                                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PURPOSE</b> | Train and evaluate 6 ML models on the 70,000-row lifestyle dataset. Select the best by AUC. This becomes Model B (lifestyle + clinical risk model).                                                                                  |
| <b>INPUTS</b>  | outputs/lifestyle/df_lifestyle_train.csv   outputs/lifestyle/df_lifestyle_test.csv                                                                                                                                                   |
| <b>OUTPUTS</b> | outputs/models/lifestyle_best_model.pkl   outputs/models/lifestyle_model_results.csv   outputs/models/lifestyle_feature_importance.csv   outputs/figures/lifestyle_roc_curves.png   outputs/figures/lifestyle_confusion_matrices.png |

### NB5 — Six models to train

| Model                  | Why Include It                                                                              | Key Hyperparameters to Tune                       |
|------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------|
| Logistic Regression    | Baseline. Interpretable. Reviewer expectation for comparison.                               | C (regularisation strength), solver               |
| Random Forest          | Handles non-linear interactions. Good on tabular data.                                      | n_estimators, max_depth, min_samples_leaf         |
| XGBoost / LightGBM     | Expected best performer. Native SHAP support. Published standard for clinical tabular data. | learning_rate, max_depth, n_estimators, subsample |
| Support Vector Machine | Strong binary classifier. Tests margin-based separation.                                    | C, kernel (RBF), gamma                            |
| k-Nearest Neighbors    | Non-parametric baseline. Shows neighbourhood structure.                                     | n_neighbors, metric                               |
| MLP (Neural Net)       | Tests whether deep representations add value over gradient boosting.                        | hidden_layer_sizes, activation, learning_rate     |

- Section 1 — Load data, separate X/y, check class balance.
- Section 2 — SMOTE on training set only if class imbalance ratio > 1.5:1.
- Section 3 — 5-fold stratified cross-validation for all six models. Record mean  $\pm$  std of: Accuracy, Precision, Recall, F1, AUC-ROC.
- Section 4 — Final evaluation on held-out test set for all six. Plot all 6 ROC curves on one figure. Plot confusion matrices on a 2x3 grid.
- Section 5 — Select best model by test set AUC. Save as lifestyle\_best\_model.pkl.
- Section 6 — Feature importance: TreeExplainer for tree models, coefficients for LR, permutation importance for SVM/MLP. Save top 15 features.

### NB6: model\_training\_clinical.ipynb

|                |                                                                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PURPOSE</b> | Same structure as NB5 but trained on the 1,190-row clinical dataset. This is the richer clinical features model. Smaller N means higher overfit risk — include GridSearchCV. |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                |                                                                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INPUTS</b>  | outputs/clinical/df_clinical_train.csv   outputs/clinical/df_clinical_test.csv                                                                                                |
| <b>OUTPUTS</b> | outputs/models/clinical_best_model.pkl   outputs/models/clinical_model_results.csv   outputs/models/clinical_feature_importance.csv   outputs/figures/clinical_roc_curves.png |

## NB6 — Key differences from NB5

- With only ~833 training rows (70% of 1190), overfitting is the main risk. Add GridSearchCV for the top 2 models from cross-validation.
- If train AUC - test AUC > 0.08, flag explicitly as overfit and apply stronger regularisation (increase min\_samples\_leaf, reduce max\_depth, add L2 regularisation).
- Consider class\_weight='balanced' as an alternative to SMOTE given the smaller N.
- The clinical model is important for PulsePhysio initialization (if used) because it has continuous clinical features. Keep track of feature importance — clinical features like trestbps, chol, and thalach that Pulse can simulate should show up as important.

| NB7: genetic_risk_integration.ipynb |                                                                                                                                                                                                                |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PURPOSE</b>                      | Integrate the population-level PRS scalar into both trained models to produce genetically-augmented risk scores. Assign patients to risk bands. This is the core novelty step.                                 |
| <b>INPUTS</b>                       | outputs/genetics/prs_population_score.csv   outputs/models/lifestyle_best_model.pkl   outputs/models/clinical_best_model.pkl   outputs/lifestyle/df_lifestyle_test.csv   outputs/clinical/df_clinical_test.csv |
| <b>OUTPUTS</b>                      | outputs/integrated/lifestyle_risk_scores_with_pgs.csv   outputs/integrated/clinical_risk_scores_with_pgs.csv   outputs/integrated/risk_stratification_bands.csv   outputs/figures/risk_stratification_plot.png |

## NB7 — Integration formula

The population PRS scalar cannot be used as an individual prediction — it represents the Indian-ancestry baseline risk contribution from genetics. The integration formula is a weighted combination:

$$p_{\text{integrated}} = w_1 \times p_{\text{model}} + w_2 \times \text{sigmoid}(\text{PRS}_{\text{population}})$$

Where:  $w_1 + w_2 = 1$ . Start with  $w_1 = 0.85$ ,  $w_2 = 0.15$  as defaults (model dominates; genetics provides a calibration nudge).

- Section 1 — Generate base model probabilities: `model.predict_proba(X_test)[: , 1]` for each test set.
- Section 2 — Compute  $p_{\text{integrated}}$  using the formula above.
- Section 3 — Sensitivity analysis: vary  $w_2$  from 0.05 to 0.40 in steps of 0.05. Show how integrated risk scores change. This goes in supplementary material.
- Section 4 — Risk banding: Low (< 0.25), Moderate (0.25-0.50), High (0.50-0.75), Very High ( $\geq 0.75$ ). Compute band distribution. Validate that higher bands have higher observed CAD prevalence.
- Section 5 — Compare AUC and Brier score:  $p_{\text{model}}$  vs  $p_{\text{integrated}}$ . Document honestly — if the genetic offset does not improve AUC, that is scientifically valid and expected (population-level offset adds calibration, not discrimination). It shifts all predictions by a fixed amount and is justified by the Indian population's systematically underestimated CAD risk in Framingham-calibrated tools.

- Section 6 — Visualisations: violin plot of integrated scores by true label, stacked bar chart of risk band distribution, calibration curve.

| NB8: fusion_and_explainability.ipynb |                                                                                                                                                                                                                                                                                         |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PURPOSE                              | Build the stacked meta-learner that fuses lifestyle and clinical model outputs into a single calibrated CAD risk score. Compute SHAP attributions with domain grouping. This is the fusion and explainability layer.                                                                    |
| INPUTS                               | outputs/models/lifestyle_best_model.pkl   outputs/models/clinical_best_model.pkl   outputs/integrated/lifestyle_risk_scores_with_pgs.csv   outputs/integrated/clinical_risk_scores_with_pgs.csv   Full feature matrices from NB1 and NB2                                                |
| OUTPUTS                              | outputs/models/fusion_meta_learner.pkl   outputs/models/fusion_calibrated_model.pkl   outputs/explainability/shap_values_lifestyle.pkl   outputs/explainability/shap_values_clinical.pkl   outputs/explainability/domain_attributions.csv   outputs/figures/shap_waterfall_examples.png |

## NB8 — Stacking implementation

Stacking works by training a meta-learner on the out-of-fold predictions of the base models. This prevents data leakage (the meta-learner never sees predictions made on data the base models were trained on).

6. Split training data into 5 folds using StratifiedKFold.
7. For each fold: train both base models on 4 folds, generate predictions on the held-out fold.
8. After all 5 folds: you have out-of-fold predictions for every training sample from both models.
9. Build a meta-feature matrix: [p\_lifestyle, p\_clinical] for each training sample.
10. Train a logistic regression meta-learner on this matrix with the true labels.
11. At inference: run new patient through both base models → get two probabilities → pass to meta-learner → get final fused probability.
12. Apply Platt scaling or isotonic regression to calibrate the final probability.

Use `sklearn.ensemble.StackingClassifier` with `cross_val_predict` internally. This handles the fold logic cleanly.

## NB8 — SHAP explainability

Apply SHAP TreeExplainer to the best lifestyle model (XGBoost/LightGBM). Compute SHAP values for the test set.

- Domain grouping: assign each feature to Genetic (PRS, family history), Lifestyle (smoke, alco, active, BMI), or Clinical (BP, cholesterol, glucose). Sum absolute SHAP values per domain per patient. Normalise to percentages.
- Output per patient: 'Genetics: X% | Lifestyle: Y% | Clinical: Z%' plus top 5 individual feature drivers.
- Visualisations: SHAP summary plot (beeswarm), SHAP waterfall plots for 3-5 example patients (high risk, low risk, borderline), SHAP bar plot for mean feature importance.
- These SHAP figures go directly into the paper as Figure 2.

## NB9: digital\_twin\_counterfactual.ipynb

|                |                                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PURPOSE</b> | The core Digital Twin module. Implements the ML-based counterfactual engine: perturbs modifiable lifestyle features in the raw feature vector, re-runs the full pipeline, computes risk deltas, and generates ranked intervention recommendations with simulated trajectory projections. |
| <b>INPUTS</b>  | outputs/models/fusion_calibrated_model.pkl   Full trained pipeline (all preprocessing + models in one sklearn Pipeline)   Patient feature vectors (from test sets)                                                                                                                       |
| <b>OUTPUTS</b> | outputs/digital_twin/intervention_results.csv (per-patient intervention rankings)   outputs/figures/trajectory_visualization.png   outputs/figures/intervention_ranking_example.png   outputs/digital_twin/sanity_check_results.csv                                                      |

### NB9 — Pre-defined intervention scenarios

| Scenario ID | Intervention                      | Feature(s) Changed                               | Published Effect (benchmark)                                       |
|-------------|-----------------------------------|--------------------------------------------------|--------------------------------------------------------------------|
| S1          | Quit smoking                      | smoke = 0                                        | ~36% relative CAD risk reduction (Critchley & Capewell, JAMA 2003) |
| S2          | Increase physical activity        | active = 1 (if inactive)                         | ~14% CAD risk reduction (Chomistek et al., Circulation 2011)       |
| S3          | Lose 5% body weight               | BMI reduced by 5%; weight reduced proportionally | ~3-5% CAD risk reduction (multiple meta-analyses)                  |
| S4          | Reduce alcohol                    | alco = 0 (if current drinker)                    | Modest BP and CV risk reduction                                    |
| S5          | Combined: quit smoking + exercise | Apply S1 and S2 simultaneously                   | Expected > either alone                                            |

### NB9 — Counterfactual engine logic

The engine is a Python function. Here is the conceptual implementation:

- Step 1: Take the patient's raw feature vector  $X_{\text{patient}}$ .
- Step 2: Compute  $\text{current\_risk} = \text{full\_pipeline.predict\_proba}(X_{\text{patient}})[1]$ .
- Step 3: For each intervention scenario  $S_i$ : copy  $X$  to  $X_{\text{modified}}$ , apply feature perturbation, compute  $\text{new\_risk} = \text{full\_pipeline.predict\_proba}(X_{\text{modified}})[1]$ , compute  $\text{delta} = \text{current\_risk} - \text{new\_risk}$ , compute  $\text{pct\_reduction} = \text{delta} / \text{current\_risk} \times 100$ .
- Step 4: Rank all scenarios by delta (descending). Output ranked recommendation list.
- Step 5: Example output: 'Quit smoking: 34% → 21% (delta -13%, relative reduction 38%). Increase exercise: 34% → 29% (delta -5%, relative reduction 15%).'

### NB9 — Trajectory visualisation

This is a VISUALISATION TOOL, not a trained temporal model. Frame it explicitly as 'simulated projection' in the paper.

- Take patient's current fused risk score at their current age.
- Use Framingham lifetime risk age-progression tables (Anderson et al., 1991) to project risk forward to ages +5 and +10 years under a 'no change' scenario.

- Apply the combined counterfactual delta to create a second 'intervention adopted' trajectory.
- Plot two diverging curves with clinical threshold lines at 7.5% and 10% risk.
- Add actual longitudinal data points if any patients have follow-up data (even a small subset strengthens validation).

## NB10 (OPTIONAL): pulse\_physio\_integration.ipynb

|         |                                                                                                                                                                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PURPOSE | [CONDITIONAL] Integrates PulsePhysio for haemodynamically-grounded intervention simulation. Only execute if scope approval requires a physiological simulator. Uses the 1,190 clinical dataset patients (continuous BP and HR available) as the Pulse patient population. |
| INPUTS  | outputs/clinical/df_clinical_test_raw.csv (unscaled continuous clinical values for Pulse initialization)   outputs/models/fusion_calibrated_model.pkl                                                                                                                     |
| OUTPUTS | outputs/pulse/pulse_haemodynamic_deltas.csv  <br>outputs/pulse/pulse_updated_risk_scores.csv  <br>outputs/figures/pulse_vs_ml_delta_comparison.png                                                                                                                        |

### WHEN TO USE NB10

Use NB10 ONLY IF the project scope requires a physiological digital twin and the ML-only counterfactual engine (NB9) is not accepted as sufficient.

If scope can be negotiated to ML-only: skip NB10 entirely. The paper will be stronger and faster to complete without it.

If NB10 is included: Pulse simulates haemodynamic responses (BP, HR, cardiac output changes) for exercise, weight loss, and smoking cessation. These physiological outputs update the clinical feature vector, which is then re-run through the ML model to compute a mechanistically-grounded risk delta. Pulse does NOT replace the ML model. It is an upstream haemodynamic simulator whose outputs feed the existing ML pipeline.

## NB10 — How PulsePhysio integrates (if used)

| Intervention                          | Pulse Mechanism                                       | Pulse Output Variable                   | Maps to ML Feature           |
|---------------------------------------|-------------------------------------------------------|-----------------------------------------|------------------------------|
| Regular aerobic exercise (150 min/wk) | ExerciseAction + chronic cardiac adaptation modifiers | Resting HR decreases; MAP decreases     | thalach (max HR), resting_bp |
| 5% weight loss                        | Modify body composition parameters                    | MAP decreases; cardiac workload reduces | resting_bp, BMI              |
| Smoking cessation                     | Increase vascular compliance                          | MAP decreases; pulse pressure decreases | resting_bp                   |

- Pulse CANNOT simulate: cholesterol changes from diet (use direct ML feature perturbation instead), or glucose/HbA1c changes. These remain ML-only counterfactuals even when Pulse is active.
- Patient initialization: take resting\_bp, max\_hr, age, sex, BMI from df\_clinical\_test\_raw.csv. Set MAP =  $(2 \times \text{diastolic} + \text{systolic}) / 3$ . Set Pulse body morphology from BMI, age, sex.
- Run each intervention scenario in Pulse, extract new haemodynamic steady-state values, replace relevant features in ML feature vector, re-run fusion model, compute risk delta.
- Validation: compare Pulse-simulated BP reduction from exercise to published values (Cornelissen & Smart, JAHA 2013: regular aerobic exercise reduces resting systolic BP by ~3.5 mmHg on average). If Pulse output is within a plausible range, the simulation is validated.

- Compare Pulse-grounded risk delta against the ML-only counterfactual delta for the same patients. If they agree within a reasonable margin, this supports the ML-only approach as valid even without Pulse. This comparison goes in the paper as supporting evidence.

| NB11: evaluation_and_paper_figures.ipynb |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PURPOSE                                  | Produces all publication-ready figures, statistical tests, master results table, and the clinical sanity check battery. Everything needed for the results and methods sections of the paper.                                                                                                                                                                                                                                                                                 |
| INPUTS                                   | All model result CSVs + all risk score CSVs + per-SNP contribution CSV + all .pkl model files                                                                                                                                                                                                                                                                                                                                                                                |
| OUTPUTS                                  | outputs/paper/master_results_table.csv   outputs/paper/master_results_table.tex  <br>outputs/paper/figure_1_model_comparison_roc.png  <br>outputs/paper/figure_2_shap_attribution.png   outputs/paper/figure_3_risk_bands.png  <br>outputs/paper/figure_4_snp_contributions.png  <br>outputs/paper/figure_5_calibration_curves.png  <br>outputs/paper/figure_6_counterfactual_examples.png  <br>outputs/paper/sanity_check_results.csv   outputs/paper/statistical_tests.csv |

## NB11 — What to produce

- Master results table: all 6 models × 2 datasets × [Accuracy, Precision, Recall, F1, AUC, Train-Test AUC Gap]. Export as LaTeX via `df.to_latex()`.
- Figure 1: Side-by-side grouped bar chart of AUC across all models and both datasets. This is the main results figure.
- Figure 2: SHAP domain attribution breakdown for a representative set of patients (high/medium/low risk). Waterfall plot for one high-risk patient.
- Figure 3: Risk band distribution (Sankey or stacked bar) by actual CVD status.
- Figure 4: Lollipop chart of per-SNP PRS contributions sorted by magnitude. Annotate top 5 by rsID and chromosomal region.
- Figure 5: Calibration curves (predicted vs observed) for base model vs integrated model vs Framingham.
- Figure 6: Counterfactual intervention results for 2-3 example patients showing current risk, scenario risks, and trajectory curves.
- Ablation table: AUC for Genetic-only, Lifestyle-only, Clinical-only, Lifestyle+Clinical, Full fused. DeLong test p-values comparing full model to each component. This is critical for publication.
- Statistical tests: McNemar's test or DeLong's test between lifestyle model and clinical model. Report p-values.
- Clinical sanity check battery: Run all 20+ scenarios listed in Section 5C below. Every single one must pass. Document as a table in supplementary material.
- Benchmark comparisons: Re-run Framingham, PCE on your test set (using published formulas or the lifelines Python library) and compare AUC, Brier score, NRI, IDI. This is the central comparison in the paper.



## 4. Complete Feature Specification

### 4.1 Feature roles across the pipeline

| Feature                                      | Dataset Source                             | Type             | Role in Model                                                                     | Simulated in DT?                                    |
|----------------------------------------------|--------------------------------------------|------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------|
| Population PRS scalar                        | PGS2809 + GenomIndia (NB4)                 | Float (scalar)   | Fixed genetic baseline offset. Added as a constant feature nudge to all patients. | NO — static                                         |
| Age                                          | Both datasets                              | Integer (years)  | Important non-modifiable risk factor.                                             | NO — static                                         |
| Sex                                          | Both datasets                              | Binary (0/1)     | Non-modifiable. Affects risk calibration.                                         | NO — static                                         |
| Systolic / Diastolic BP                      | 1190 clinical                              | Float (mmHg)     | Clinical feature. Responds to exercise/weight loss.                               | INDIRECT — via Pulse (if NB10) or proxy             |
| Total cholesterol                            | 1190 clinical (mg/dL); 70k (ordinal 1/2/3) | Float or Ordinal | Strong CAD predictor.                                                             | PARTIAL — ML perturbation for dietary changes       |
| Max HR (thalach)                             | 1190 clinical                              | Float (bpm)      | Cardiorespiratory fitness proxy. Changes with exercise.                           | INDIRECT — via Pulse                                |
| BMI                                          | 70k (derived)                              | Float (kg/m2)    | Bridges lifestyle and clinical. Modifiable.                                       | YES — 5% reduction scenario                         |
| Smoke                                        | 70k (binary)                               | Binary (0/1)     | Modifiable lifestyle factor.                                                      | YES — set to 0                                      |
| Alcohol                                      | 70k (binary)                               | Binary (0/1)     | Modifiable lifestyle factor.                                                      | YES — set to 0 if drinking                          |
| Active (physical activity)                   | 70k (binary)                               | Binary (0/1)     | Modifiable lifestyle factor.                                                      | YES — set to 1 if inactive                          |
| Cholesterol category                         | 70k (ordinal 1/2/3)                        | Ordinal          | Used in 70k model. Note data limitation.                                          | NO (binary in 70k; use 1190 mg/dL for perturbation) |
| Fasting BS / glucose                         | 1190 clinical; 70k (ordinal)               | Binary / ordinal | Metabolic risk factor.                                                            | NO in current scope — future                        |
| Exercise angina, oldpeak, ST slope, ca, thal | 1190 clinical                              | Mixed            | Clinical ECG and angiographic features. Not modifiable.                           | NO — static                                         |

### 4.2 What gets perturbed in the counterfactual engine

#### Simulation Space Definition

The DT ONLY perturbs features that a patient can actually change through lifestyle intervention.

MODIFIABLE (direct simulation targets): smoke, alco, active, BMI.

INDIRECTLY MODIFIABLE (via Pulse if NB10, or via direct perturbation for sensitivity): resting\_bp, thalach.

FIXED (never perturbed): age, sex, PRS scalar, all ECG/angiographic features, cholesterol category in 70k model (too coarse for meaningful perturbation).

HYBRID (perturbed via direct feature change for dietary scenarios): cholesterol value in 1190 model (e.g., set chol from 240 to 200 mg/dL to simulate dietary improvement).



## 5. Full Validation Strategy — Publication Grade

Validation is the most critical section for publication. A paper with good methodology but weak validation will not be accepted. Every component must be validated separately.

### 5A. ML Risk Model Validation

| Metric                                      | What It Measures                                                      | Target                                                                     | How to Compute                                                                                                             |
|---------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| AUC-ROC                                     | Discrimination — how well the model separates cases from controls     | > 0.78 on held-out test set                                                | <code>sklearn.metrics.roc_auc_score</code> on test set; compare to Framingham AUC using DeLong test ( $p < 0.05$ required) |
| Brier Score                                 | Calibration — reliability of predicted probabilities                  | < 0.15                                                                     | <code>sklearn.metrics.brier_score_loss</code> ; plot calibration curve (predicted vs observed in risk deciles)             |
| Sensitivity / Specificity                   | Clinical relevance at decision thresholds (7.5% and 10%)              | Sensitivity > 0.75 at 10% threshold                                        | Compute at each clinical threshold; report in results table                                                                |
| NRI (Net Reclassification Index)            | Proportion of patients correctly reclassified vs Framingham           | Positive NRI                                                               | Category-based NRI vs Framingham using same Indian test set                                                                |
| IDI (Integrated Discrimination Improvement) | Average improvement in predicted probability separation vs Framingham | Positive IDI                                                               | Report 95% CI via bootstrap; use <code>nricens</code> or <code>sklearn-compatible</code> library                           |
| Decision Curve Analysis                     | Net clinical benefit across risk thresholds                           | Model should show higher net benefit than Framingham for Indian population | <code>dcurves</code> Python package or <code>R dca()</code>                                                                |

### 5B. Fusion Layer Ablation Study

This is mandatory for publication. You must show that combining all three sources (genetic + lifestyle + clinical) outperforms any individual component. Report as Table in Results section.

| Model Condition                  | Features Used                                            | Expected AUC                | Purpose                                                      |
|----------------------------------|----------------------------------------------------------|-----------------------------|--------------------------------------------------------------|
| Genetic only                     | PRS scalar only (logistic regression)                    | ~0.55-0.62                  | Shows genetic risk alone is insufficient                     |
| Lifestyle only (Model B, no PRS) | 70k features excluding PRS                               | ~0.72-0.76                  | Baseline lifestyle model performance                         |
| Clinical only (Model B on 1190)  | 1190 clinical features                                   | ~0.74-0.78                  | Baseline clinical model performance                          |
| Full Model B (with PRS)          | All features including PRS offset                        | ~0.75-0.79                  | Effect of genetic integration on ML model                    |
| Framingham (benchmark)           | Standard Framingham variables (applied to same test set) | ~0.68-0.72 on Indian cohort | External benchmark — expected to underperform on Indian data |

|                                |                                              |                 |                                                          |
|--------------------------------|----------------------------------------------|-----------------|----------------------------------------------------------|
| STACKED FUSION<br>(our system) | Meta-learner on Model A +<br>Model B outputs | > 0.80 (target) | Full system — must significantly<br>outperform all above |
|--------------------------------|----------------------------------------------|-----------------|----------------------------------------------------------|

Use DeLong's test to compare AUCs between the fused model and each individual component. All comparisons must be statistically significant ( $p < 0.05$ ).

### 5C. PRS Layer Validation

- Divide dataset into PRS tertiles (low / medium / high) using Indian population reference distribution.
- Compute observed CAD event rate in each tertile. Show dose-response: Tertile 3 > Tertile 1. Report odds ratio with 95% CI.
- Compare Indian-calibrated PRS stratification against European-calibrated PRS (re-run with European reference frequencies). Show that Indian calibration provides better stratification in your cohort. This directly justifies the GenomeIndia contribution.

### 5D. Digital Twin / Counterfactual Engine Validation

This is the hardest component to validate and the one most likely to be challenged. Use all three methods.

#### Method 1 — Clinical Literature Benchmarking (PRIMARY, mandatory)

Your DT predicts a risk reduction from each lifestyle intervention. Compare these predictions against published RCT effect sizes. If your predictions fall within the 95% CI of the literature, that is validation.

| Intervention                 | Published Effect Size            | Source                             | Your DT Must Predict             |
|------------------------------|----------------------------------|------------------------------------|----------------------------------|
| Smoking cessation            | ~36% relative CAD risk reduction | Critchley & Capewell, JAMA 2003    | 20% to 45% range                 |
| Exercise (150 min/wk)        | ~14% CAD risk reduction          | Chomistek et al., Circulation 2011 | 8% to 20% range                  |
| 5% weight loss               | ~3-5% CAD risk reduction         | Multiple meta-analyses             | 2% to 7% range                   |
| Mediterranean diet adherence | ~30% CVD event reduction         | PREDIMED trial, NEJM 2013          | 15% to 40% range                 |
| Combined smoking + exercise  | > either alone                   | Multiple sources                   | Must exceed either solo scenario |

#### Method 2 — Clinical Sanity Check Battery (mandatory — all must pass)

Run 20+ predefined scenarios. Every single one must produce the expected directional result. Report as supplementary table.

| Sanity Test            | Feature Change | Expected Direction | Pass/Fail |
|------------------------|----------------|--------------------|-----------|
| Smoker vs never-smoker | smoke: 1 → 0   | Risk DECREASES     | [ ]       |
| Sedentary → active     | active: 0 → 1  | Risk DECREASES     | [ ]       |
| Healthy BMI → obese    | BMI: 22 → 32   | Risk INCREASES     | [ ]       |

|                                            |                                |                                                     |     |
|--------------------------------------------|--------------------------------|-----------------------------------------------------|-----|
| Normal LDL → high LDL (1190 model)         | chol: 160 → 240 mg/dL          | Risk INCREASES                                      | [ ] |
| Low HDL → high HDL (1190 model)            | HDL analogue via chol proxy    | Risk DECREASES                                      | [ ] |
| Normal BP → hypertensive (1190 model)      | trestbps: 120 → 160 mmHg       | Risk INCREASES                                      | [ ] |
| Normal HbA1c → diabetic range (1190 model) | fbs: 0 → 1                     | Risk INCREASES                                      | [ ] |
| Low PRS → high PRS                         | PRS scalar: low → high tier    | Risk INCREASES                                      | [ ] |
| Combined: quit smoking + exercise          | Apply S1 and S2 simultaneously | Risk lower than either alone                        | [ ] |
| Age increase +10 years (same features)     | age: 45 → 55                   | Risk INCREASES                                      | [ ] |
| Male vs female (same features)             | sex: 1 → 0                     | Risk should differ (male typically higher baseline) | [ ] |
| Quit drinking (if drinker)                 | alco: 1 → 0                    | Risk DECREASES                                      | [ ] |

### Method 3 — Longitudinal Validation (if any follow-up data exists)

If any patients in your datasets have follow-up measurements after a documented lifestyle change, validate that your model's predicted risk delta matches the direction and approximate magnitude of the observed risk change. Even n=30-50 patients is sufficient. Compute Pearson correlation between predicted delta and observed delta.

## 6. Execution Order and Dependencies

Follow this exact execution order. NB1 and NB2 can run in parallel. NB5 and NB6 can run in parallel after their prerequisites. NB10 is only executed if PulsePhysio is in scope.

| Step          | Notebook                                  | Dependencies                         | Can Parallelise With |
|---------------|-------------------------------------------|--------------------------------------|----------------------|
| 1             | NB1 — preprocessing_70k.ipynb             | Cardio_Data.csv (raw)                | NB2, NB3             |
| 2             | NB2 — preprocessing_1190_clinical.ipynb   | Cleveland + Hungarian + Statlog CSVs | NB1, NB3             |
| 3             | NB3 — genome_data_preprocessing.ipynb     | PGS2809 TSV + GenomeIndia TSVs       | NB1, NB2             |
| 4             | NB4 — prs_score_computation.ipynb         | NB3 must complete first              | Nothing (sequential) |
| 5             | NB5 — model_training_lifestyle.ipynb      | NB1 must complete first              | NB6                  |
| 6             | NB6 — model_training_clinical.ipynb       | NB2 must complete first              | NB5                  |
| 7             | NB7 — genetic_risk_integration.ipynb      | NB4 + NB5 + NB6 all complete         | Nothing              |
| 8             | NB8 — fusion_and_explainability.ipynb     | NB7 complete                         | Nothing              |
| 9             | NB9 — digital_twin_counterfactual.ipynb   | NB8 complete                         | NB10 (if active)     |
| 10 (optional) | NB10 — pulse_physio_integration.ipynb     | NB8 complete + NB2 raw CSV           | NB9                  |
| 11            | NB11 — evaluation_and_paper_figures.ipynb | NB9 (and NB10 if active) complete    | Nothing              |

### Recommended execution plan on Colab/Kaggle

Week 1: Run NB1, NB2, NB3 simultaneously (preprocessing). Run NB4 after NB3.  
Week 2: Run NB5 and NB6 simultaneously (model training). This is the most compute-intensive step — use GPU runtime where available for XGBoost/LightGBM on the 70k dataset.  
Week 3: Run NB7 (integration), NB8 (fusion + SHAP). SHAP computation on 70k samples can be slow — sample 1,000-5,000 test patients for SHAP if full computation times out.  
Week 4: Run NB9 (DT counterfactual engine), NB10 (optional Pulse), NB11 (evaluation + figures).  
Week 5: Paper writing using NB11 outputs directly.

# 7. Paper Framing — Exact Language to Use

These are the exact phrases and framings to use throughout the paper. Deviating from these invites reviewer rejection.

|            |                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| DO say     | Feature attribution to quantify the relative contribution of genetic, lifestyle, and clinical risk factors               |
| DO say     | Counterfactual intervention analysis to estimate personalised risk reduction under specified lifestyle modifications     |
| DO say     | Simulated risk projection using Framingham age-progression hazard functions                                              |
| DO say     | Population-level polygenic risk score calibrated to South Asian ancestry using GenomeIndia allele frequencies            |
| DO say     | ML-based digital twin following the paradigm established in Subramanian et al. (npj Digital Medicine, 2020)              |
| DO say     | Associational risk attribution with counterfactual risk projections under intervention scenarios                         |
| DO NOT say | 'Causal' (without Mendelian Randomisation machinery) — reviewers will reject immediately                                 |
| DO NOT say | 'Time-based trajectory model' unless you have longitudinal training data (you do not)                                    |
| DO NOT say | 'Simulate' when you mean 'perturb features and re-run inference' — say 'counterfactual analysis' or 'what-if estimation' |
| DO NOT say | 'Our PRS represents individual genetic risk' — it represents population-level expected genetic risk                      |
| DO NOT say | BioGears or PulsePhysio are required for a digital twin — ML-based DTs are valid in the literature                       |

## 7.1 Justifying the Digital Twin label without a physiological simulator

**Exact justification paragraph for the Methods section**

We implement a personalised digital twin as a patient-specific probabilistic risk model coupled with a counterfactual intervention engine. The twin maintains a dynamic representation of each patient's risk state as a function of their genetic, lifestyle, and clinical profile, and enables what-if simulation of lifestyle interventions by propagating feature-level perturbations through the learned risk model. This approach follows the ML-based digital twin paradigm established in Subramanian et al. (2020) and is appropriate for our goal of risk projection and intervention simulation rather than organ-level physiological modelling.

## 8. Open Questions and Things Not Yet Answered

These are items that were not fully resolved and will block progress or paper quality if left unaddressed.

| Question                                                         | Why It Matters                                                                                                                                                                                                                                                                                | Recommended Resolution                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is PulsePhysio in or out of scope?                               | NB10 adds 2-3 weeks of work. If not required, removing it makes the project faster and cleaner.                                                                                                                                                                                               | Negotiate with supervisor NOW. If ML-only DT is acceptable: drop NB10. If physiological simulation is mandatory: use Pulse, not BioGears.                                                                                                                                                                                        |
| Which Indian cohort risk score exists for comparison?            | Adding an India-specific baseline (e.g., IndiHeart, CURES score) makes the paper significantly stronger than comparing only to Framingham.                                                                                                                                                    | Search PubMed for 'Indian cardiovascular risk score' and 'South Asian CAD risk calculator'. Add whichever is most recent and validated as a fourth baseline.                                                                                                                                                                     |
| How to handle the 70k vs 1190 feature schema mismatch in fusion? | The fusion meta-learner takes outputs from both models as probabilities — so the raw feature mismatch does not directly matter for the stacking step. But you need to decide: is the fused output the average of both model predictions on a shared test set, or are they separate pipelines? | Recommended: treat them as two separate but complementary evaluation pipelines. The lifestyle model (70k) gives one AUC, the clinical model (1190) gives another, the fusion report in the paper acknowledges both. For a true single fused score, you need a dataset that has BOTH lifestyle AND clinical features per patient. |
| Do we have any Indian patients in our clinical dataset?          | Cleveland and Hungarian datasets are from Western populations. If the 1190 clinical model is validated on non-Indian patients, this must be stated as a limitation.                                                                                                                           | State explicitly in the paper: 'Clinical model validated on Western cohort data. India-specific calibration achieved through PRS integration with GenomeIndia frequencies. Future work should validate on Indian clinical cohorts (e.g., AIIMS or ICMR datasets).'                                                               |
| What is the target journal?                                      | Knowing the journal determines word limits, figure counts, and whether supplementary material is allowed for the full sanity battery.                                                                                                                                                         | Decide before writing. Suggested targets: npj Digital Medicine, Journal of Medical Internet Research, Frontiers in Cardiovascular Medicine, or IEEE Journal of Biomedical and Health Informatics.                                                                                                                                |
| SHAP computation time on 70k samples                             | SHAP TreeExplainer on 70,000 test samples can take 10-30 minutes on Colab. If it times out, the explainability step fails.                                                                                                                                                                    | Compute SHAP on a random stratified subsample of 2,000-5,000 test patients. State this in the methods. Results are statistically identical at this sample size.                                                                                                                                                                  |

## 9. Master Decision Summary

| Decision Point          | Final Answer                                                                                                                                                                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Digital twin type       | ML-based counterfactual engine (mandatory). PulsePhysio haemodynamic layer (optional/conditional on scope).                                                                                                                                                     |
| Physiological simulator | PulsePhysio ONLY if required. BioGears is excluded — wrong tool for this use case.                                                                                                                                                                              |
| Fusion method           | Stacked meta-learner (logistic regression on out-of-fold predictions). NOT fixed weighted average.                                                                                                                                                              |
| Simulation input level  | Raw features only. Perturb lifestyle features in raw vector, re-run full pipeline.                                                                                                                                                                              |
| PRS approach            | Path A: Population-level expected PRS using PGS2809 weights + GenomeIndia Indian-ancestry allele frequencies. Monte Carlo uncertainty quantification (10,000 iterations, Beta-distributed per SNP) produces 95% CI. Mean used downstream; CI reported in paper. |
| Explainability          | SHAP TreeExplainer with domain grouping (Genetic / Lifestyle / Clinical). NOT physiological simulation outputs.                                                                                                                                                 |
| Causal framing          | Never use 'causal'. Use 'feature attribution' and 'counterfactual intervention analysis'.                                                                                                                                                                       |
| Trajectory              | Simulated projection using Framingham age-progression curves. NOT a trained temporal model (no longitudinal data).                                                                                                                                              |
| Feedback loop           | Longitudinal data update protocol (store timestamped records, re-run pipeline on new data). NOT a computational feedback loop.                                                                                                                                  |
| Validation — model      | AUC, Brier score, NRI vs Framingham, IDI vs Framingham, DeLong tests, ablation study.                                                                                                                                                                           |
| Validation — DT         | Literature benchmarking against RCT effect sizes + 20+ clinical sanity checks (all must pass).                                                                                                                                                                  |
| Number of notebooks     | 10 total (NB1-NB9 + NB11 mandatory; NB10 conditional). Run NB1/NB2/NB3 in parallel; NB5/NB6 in parallel.                                                                                                                                                        |
| Primary target user     | Preventive health clinics and corporate wellness companies (B2B). NOT cardiologists at launch.                                                                                                                                                                  |
| Product being built     | Genetically-informed CAD risk intelligence report: India-calibrated 10-year risk + domain attribution + ranked intervention plan with quantified risk reductions.                                                                                               |